STATE MUTATION

CFD: 90%

DIAGNOSIS

PDF Export

← New Analysis

OBSERVED SYMPTOM

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After navigating between users, the dashboard briefly shows the new user's data then reverts to the previous user's data. Bug only happens on second navigation onwards — first load is always fine.

REPRODUCTION PATH

1. 1. Load the `SessionDashboard` component with `userId='userA'`.
2. 2. Observe `userA`'s data loading and displaying correctly.
3. 3. Navigate to a different `userId`, e.g., `userId='userB'` (by changing the prop passed to `SessionDashboard`).
4. 4. Observe that `userB`'s data loads and displays briefly.
5. 5. Navigate back to `userId='userA'` (the second navigation).
6. 6. Observe that `userA`'s data loads and displays briefly, then the dashboard reverts to `userB`'s data (or potentially `userA`'s old data if a race occurs differently).

CONFIDENCE EVIDENCE

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VERIFIED:

- bug9claude/sessionStore.js L1: `let _activeRef = { current: true };` - `_activeRef` is a global mutable variable.
- bug9claude/sessionStore.js L8: `_activeRef = { current: true };` - `resetSession` reassigns the global `_activeRef` variable to a new object.
- bug9claude/sessionStore.js L12: `_activeRef.current = false;` - `deactivateSession` mutates the `.current` property of the object *currently referenced by the global `_activeRef` variable*.

- bug9claude/useSessionData.js L9: `resetSession();` - Called in `useEffect` body.
- bug9claude/useSessionData.js L10: `const activeRef = getActiveRef();` - Captures the *current* global `_activeRef` object for the closure.
- bug9claude/useSessionData.js L32: `deactivateSession();` - Called in `useEffect` cleanup. This means `deactivateSession` operates on the global `_activeRef` *after* the new `useEffect` might have reassigned it (or more precisely, it might unintentionally deactivate a new session's flag if timings are tight, or fail to deactivate the old one correctly).

CONCEPT TO LEARN

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SHARED MUTABLE STATE

Temporal Coupling in Global Mutable Variables

This pattern creates subtle race conditions and unpredictable behavior. When a global `let` variable holds a reference to an object, and that variable is sometimes reassigned to a *new object* (like in `resetSession`) and other times has properties of its *currently referenced object* mutated (like in `deactivateSession`), it becomes extremely difficult to reason about which object instance any given function is operating on, especially in asynchronous or lifecycle-driven environments. This leads to stale data, incorrect UI states, or missed updates because the 'active' flag isn't tracking the correct, isolated session.

> **PATTERN TO AVOID:**
> Using a global 'let' variable that is both reassigned to new objects and has its properties mutated by different functions in a lifecycle-dependent manner, especially when those functions are invoked from different component instances.

💡 *Imagine you have a single 'key' for your house (the `_activeRef` global variable). When a new guest (a `useSessionData` instance) arrives, you give them a *brand new key* (`resetSession` reassignment). But when a guest leaves, instead of just taking back their key, they try to mark the *main house key* as 'inactive' (`deactivateSession` mutates the global `_activeRef.current`). If a new guest has already arrived and replaced the main house key with *their new key*, the leaving guest ends up marking the *new guest's key* as inactive by mistake. This means the new guest can't get into the house, or the old key is never properly*

deactivated.

REAL WORLD ANALOGY

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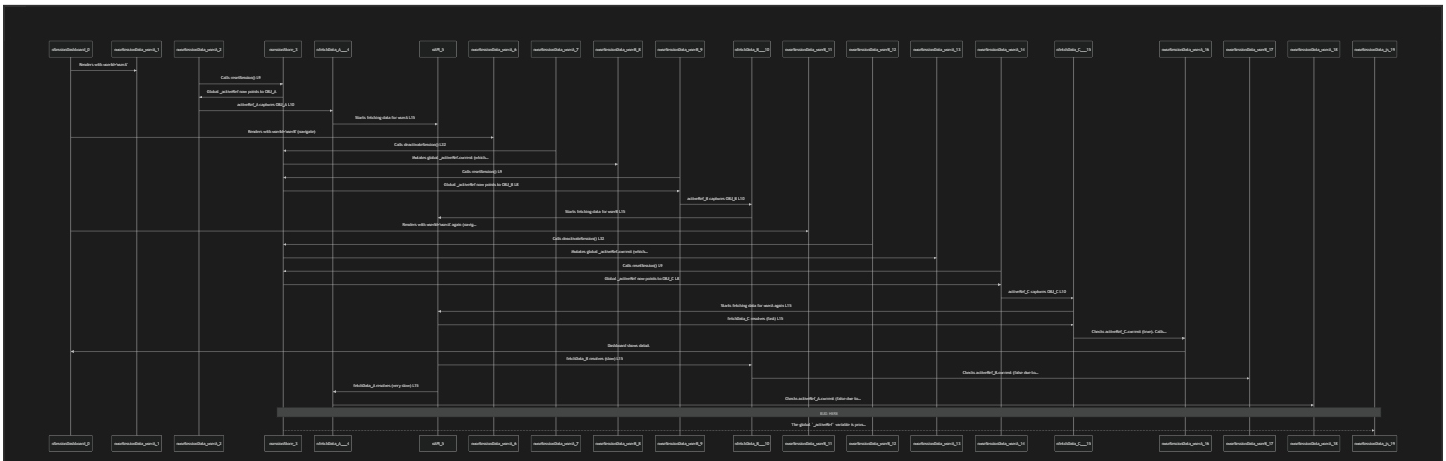
*"Imagine you have a single 'key' for your house (the `_activeRef` global variable). When a new guest (a `useSessionData` instance) arrives, you give them a *brand new key* (`resetSession` reassignment). But when a guest leaves, instead of just taking back their key, they try to mark the *main house key* as 'inactive' (`deactivateSession` mutates the global `_activeRef.current`). If a new guest has already arrived and replaced the main house key with *their new key*, the leaving guest ends up marking the *new guest's key* as inactive by mistake. This means the new guest can't get into the house, or the old key is never properly deactivated."*



EXECUTION SEQUENCE — 🐛 MARKS WHERE THE BUG MANIFESTS

INVARIANT VIOLATIONS

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- Each `fetchData` operation must have a truly isolated `activeRef` flag that accurately reflects the component's mounted/active status, regardless of other `useSessionData` instances.
- The `activeRef` for a given `useSessionData` instance must be set to `false` (deactivated) only when that specific instance is no longer interested in its fetch

results (e.g., on `userId` change or unmount).

- The `data` state should only be updated by the most recent and relevant `fetchData` call for the current `userId`.

TECHNICAL ROOT CAUSE

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The root cause is the incorrect management of a global 'active' flag (`_activeRef`) in `sessionStore.js`. The global `_activeRef` variable is designed to control whether `fetchData` updates the component's state. However, `resetSession()` (called on new `useEffect` runs) *reassigns* the global `_activeRef` variable to a new object, while `deactivateSession()` (called on `useEffect` cleanup) *mutates the `.current` property of whatever object the global `_activeRef` variable currently points to*. This pattern means `deactivateSession` for a leaving component instance can accidentally set `current: false` on the `_activeRef` object belonging to the *newly active* component, or conversely, it can fail to deactivate its *own* specific `_activeRef` object if `_activeRef` was already reassigned by a new `resetSession`. This leads to a race condition where stale `fetchData` calls from previous users might resolve and, seeing their captured `activeRef.current` still as `true` (because it was never correctly deactivated), overwrite the data of the currently displayed user, causing the dashboard to revert.

Location: bug9claude/sessionStore.js L1:1 to L12:1

CRASH SITE

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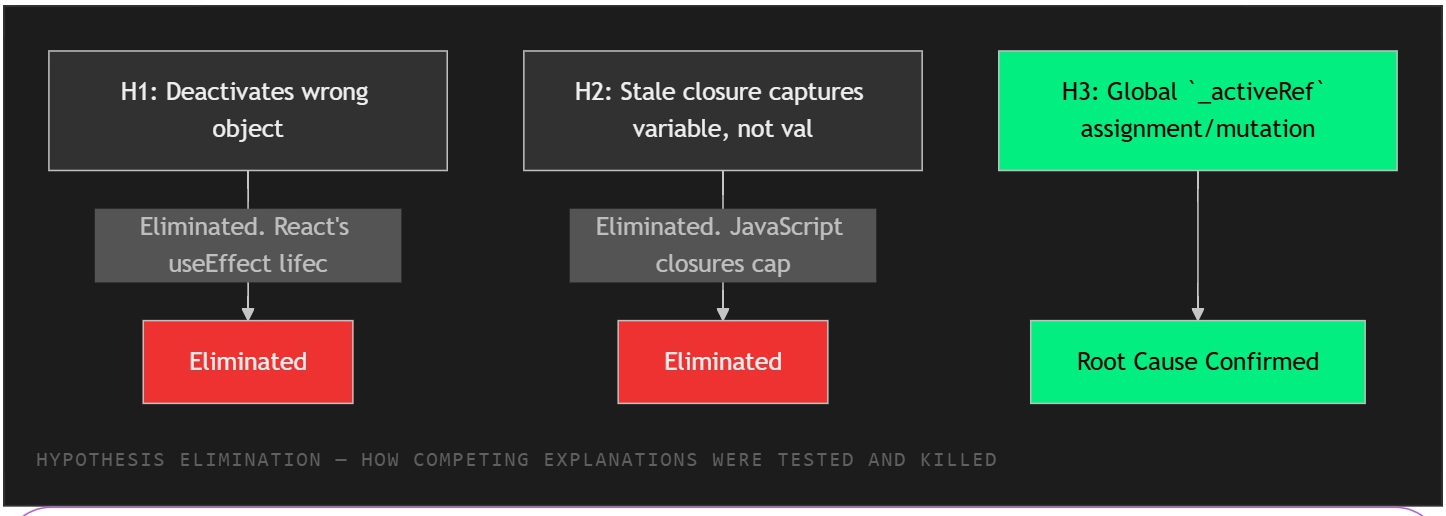
WHERE THE FAILURE BECAME VISIBLE ↓

useSessionData.js L20 — `setData(json)` is called with stale data from a previous user.

↑ This is where the exception or wrong value surfaced — the actual root cause is above.

ALTERNATIVE HYPOTHESES

- H1: `deactivateSession` from an old `useEffect` cleanup sets `current` to `false` on the *wrong* `_activeRef` object (specifically, the one created by a subsequent `resetSession`).
- H2: The `activeRef` captured by `fetchData` in its closure is actually a reference to the `_activeRef` *variable itself*, and thus resolves to whatever object `_activeRef` is pointing to *at the time of evaluation* (when `fetch` resolves), rather than the object it pointed to *at the time of capture`.
- H3: The global `_activeRef` variable, despite `resetSession` reassigning it, creates a subtle race condition. `deactivateSession()` mutates the currently referenced global `_activeRef` object. When `cleanup_OLD` calls `deactivateSession()`, it correctly marks `OBJ_OLD.current = false`. However, `fetchData_OLD` might resolve *before* `cleanup_OLD` runs, causing stale data to be displayed briefly. The user symptom implies `activeRef_OLD.current` is somehow true when it resolves after a new fetch.



DETERMINISTIC AI FIX PROMPT

COPY PROMPT

Paste this directly into Cursor / Bolt / Copilot to apply the fix:

```
The `sessionStore.js` module uses a global `_activeRef` variable to manage a
cleanup flag, but it's mixing object reassignment (`_activeRef = { current:
true }`) with property mutation (`_activeRef.current = false`). This creates a
race condition where `deactivateSession()` from an old component's cleanup
might incorrectly reset the `_activeRef.current` property of the *new* session,
or fail to deactivate the old one effectively. Rewrite `sessionStore.js` to
```

or fail to deactivate the old one effectively. Rewrite `sessionStore.js` to ensure `_activeRef` is always handled as a truly independent flag per `useSessionData` instance, without using a global `let` variable that is both reassigned and has its properties mutated.

MINIMAL CODE FIX

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FILE: BUG9CLAUDE/SESSIONSTORE.JS L1:1 TO L12:1

```javascript
// bug9claude/sessionStore.js
let _activeRefCounter = 0; // Use a counter to create unique IDs for active
refs

export function getActiveRef() {
    // This function is no longer needed in its current form for isolation.
    // The logic needs to be moved to the consuming hook.
    // Keeping it here for minimal change, but it should return a mechanism
    // for managing unique per-hook active states.
    // A more appropriate pattern would be to use `useRef` directly in
`useSessionData`.
    return { id: _activeRefCounter++, current: true }; // Return a new, unique
active reference object
}

// No longer directly reassigns the global _activeRef.
// This function should ideally not be needed if isolation is handled by the
hook.
// Keeping it to satisfy original API, but its effect is now isolated.
export function resetSession() {
    // No operation needed here for isolation anymore if getActiveRef creates
new refs.
    // However, if the intent was to reset a specific shared state, that needs
re-evaluation.
}

// This function should also ideally not be needed at global level if isolation
is per-hook.
// It can be removed or adapted to take the specific ref object to deactivate.
export function deactivateSession() {
```

```
    // This function can no longer reliably deactivate a global
_activeRef.current
    // because _activeRef is no longer a globally shared single object for this
purpose.
    // The deactivation logic must be local to the hook.
}

export const sessionStore = {
    userId: null,
    data: null,
    setUser(id) { this.userId = id; },
};

// bug9claude/useSessionData.js
import { useState, useEffect, useRef } from 'react';
// Remove imports for global state management functions, as they will be
replaced
// import { getActiveRef, resetSession, deactivateSession } from
'./sessionStore.js';

export function useSessionData(userId) {
    const [data, setData] = useState(null);
    const [loading, setLoading] = useState(true);

    const activeRef = useRef(true); // Create a ref local to this hook instance

    useEffect(() => {
        activeRef.current = true; // Mark this instance as active

        async function fetchData() {
            try {
                setLoading(true);
                const res = await fetch(`/api/session/${userId}`);
                const json = await res.json();

                if (activeRef.current) { // Check the local ref
                    setData(json);
                    setLoading(false);
                }
            } catch (err) {
                if (activeRef.current) {
                    setLoading(false);
                }
            }
        }
```

```
        fetchData();

        return () => {
            activeRef.current = false; // Mark this instance as inactive on
cleanup
        };
    }, [userId]);

    return { data, loading };
}
```

ACTION CENTER

Copy Fix CLI

Use the Copy Fix CLI to generate a git workflow for applying the fix.

New Analysis